

Project Proposal: Real-Time PEME Monitoring & Result Access System

Prepared for: American Hospital Inc. (AHI), Intramuros, Manila

1. Executive Summary

The Real-Time PEME (Pre-Employment Medical Examination) Monitoring and Result Access System is a custom-built, modern web application designed to eliminate paper-based tracking bottlenecks, reduce patient waiting times, and automate the collation of medical results.

By transitioning from a manual routing system to an active digital tracking platform, this system provides AHI staff with live visibility of patient flow across all 10 departments. Furthermore, it includes secure, mobile-friendly portals for patients and recruitment agencies, drastically reducing phone and email status inquiries while ensuring strict compliance with the Data Privacy Act (RA 10173).

2. Core System Features & Operational Benefits

Real-Time Departmental Queuing & Monitoring

- **What it does:** Replaces paper routing slips with a live digital dashboard. When a patient finishes an exam (e.g., X-Ray), the department staff clicks "Complete," and the patient instantly appears on the next department's dashboard.
- **Business Value:** Eliminates the "blind spots" where patients get lost in the clinic. Reception and Nurse Coordinators can see exactly where every patient is, directly targeting the reduction of the current 2.6-hour wait time and preventing bottleneck buildups in high-traffic areas like the Laboratory.

Active Clinical Data Encoding (Specialized EMR)

- **What it does:** Allows clinical staff to encode specific PEME findings directly into the system at the point of care.

- **Business Value:** Completely eliminates the 224 monthly manual collation incidents and 160 encoding errors. Completed results will no longer sit idle for 1 to 3 hours waiting for manual retrieval by releasing staff.

Automated Result Collation & Physician Dashboard

- **What it does:** Automatically gathers all encoded test results for a specific patient and presents them to the Medical Director in a clean, consolidated view to make a "Fit to Work" or "Unfit" decision.
- **Business Value:** Saves the Releasing Staff hours of manual data compilation and re-typing, directly addressing the 72 monthly failures to meet the 72-hour regular release targets and 11 monthly failures for 24-hour rush targets.

Secure Agency & Employer Portal

- **What it does:** Provides recruitment agencies with a secure login to view the real-time status of *only* their assigned applicants.
- **Business Value:** Drastically reduces the 50 to 65 weekly status inquiry calls the clinic currently receives. Agencies can independently check statuses and download finalized PEME certificates once released.

Mobile-Ready Patient Portal & Automated Email Alerts

- **What it does:** Allows individual applicants to log in via their smartphones to track their progress and automatically emails them when their final medical certificate is ready.
- **Business Value:** Enhances the patient experience by providing transparency, reducing anxiety, and minimizing questions directed at the reception staff.

"Read-Only" Financial Integration

- **What it does:** The system operates strictly as a clinical workflow tool. It does not manage or modify any financial or billing data. It simply checks if a patient is marked "Paid" before allowing exams to proceed.
 - **Business Value:** Ensures absolute safety and integrity of AHI's legacy financial and admission systems.
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3. The Technology Stack (How it Works)

The system is built using enterprise-grade, modern web technologies designed for speed, security, and reliability.

- **The User Interface (React.js / Next.js):** This is the technology used to build the visual dashboards. It creates a highly responsive "Single Page Application." This means dashboards load instantly (under 2 seconds) and adapt flawlessly to any screen size—from the large desktop monitors used by the Medical Director to the smartphones used by patients.
 - **The Secure Data Vault (Supabase / PostgreSQL):** Supabase acts as the secure, cloud-based vault for all PEME records. Its standout feature is "**Real-Time WebSockets**," which allows the database to instantly push updates to the screen (like a live chat app). This enables the Nurse Coordinator to see a patient's status change to "Completed" the exact second the X-Ray tech finishes their work, without ever hitting the refresh button.
 - **Security & Data Privacy Compliance:** To ensure strict compliance with the **Philippine Data Privacy Act of 2012 (RA 10173)** and DOH standards, the system uses built-in Row Level Security (RLS). This guarantees that an agency representative can *only* see data belonging to their specific applicants. Furthermore, all medical data is fully encrypted.
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4. Projected Operating Costs & Return on Investment (ROI)

Because this is being developed as an academic capstone project, **all software development, coding, and design labor are provided at absolutely no cost to AHI.**

However, to support AHI's specific operational volume (1,000 patients per month, 50,000+ historical records, and 45 simultaneous staff users) with 99.9% uptime, we recommend utilizing professional cloud infrastructure. The costs for running this enterprise-grade system are highly accessible:

Estimated Monthly Infrastructure Costs (Production Environment):

- **Secure Cloud Database (Supabase Pro Plan): ~\$25 / month (approx. PHP 1,400)**
 - *Context:* This plan supports up to 100,000 monthly active users and 8GB of database storage. Since AHI handles 1,000 PEME cases a month (roughly ~1,100 total users including staff and agencies), this tier provides massive room for growth over the next 5–10 years without requiring upgrades.

- **Web Hosting & Deployment (Vercel Pro Plan): ~\$20 / month (approx. PHP 1,120)**
 - *Context:* Ensures lightning-fast loading speeds for staff dashboards and high-security encryption (HTTPS) for the external portals.
- **Custom Hospital Web Domain: ~\$15 / year (approx. PHP 840)**
 - *Context:* A custom URL (e.g., results.americanhospital.ph) for professional branding.

Total Estimated Operating Cost: ~PHP 2,520 per month (approx. PHP 31,000 annually).